

1.) a) $T(n) = 3T(\sqrt{n}) + \theta(n^2)$

Recurrence	# of nodes	operation	cost
$T(n)$	$3^0 = 1$	n^2	n^2
$T(\sqrt{n})$	$3^1 = 3$	$(\sqrt{n})^2$	$3n$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$T(\sqrt{n}^i)$	3^i	$(\sqrt{n}^i)^2$	$3n^i$

$$\sum_{i=0}^{\log_2 n} 3^i = \theta(n^2)$$

b) $T(n) = 4T\left(\frac{n}{2} + 2\right) + n$

Recurrence	# of nodes	operation	cost
$T(n)$	$4^0 = 1$	n	n
$T\left(\frac{n}{2}\right)$	$4^1 = 4$	$\left(\frac{n}{2}\right)\left(\frac{n}{2}\right)\left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$	$4\left(\frac{n}{2}\right) = 2n$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$T\left(\frac{n}{2}^i\right)$	4^i	$\left(\frac{n}{2}\right)^i$	$4^i \left(\frac{n}{2}\right)^i$

$$\sum_{i=0}^{\log_2 n} 4^i \left(\frac{n}{2}\right)^i = \theta(n^2)$$

c) $T(n) = 3T\left(\frac{n}{2}\right) + n$

Recurrence	# of nodes	operation	cost
$T(n)$	$3^0 = 1$	n	n
$T\left(\frac{n}{2}\right)$	$3^1 = 3$	$\frac{n}{2}$	$\frac{3n}{2}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$T\left(\frac{n}{2}^i\right)$	3^i	$\left(\frac{n}{2}\right)^i$	$3^i \left(\frac{n}{2}\right)^i$

$$\sum_{i=0}^{\log_2 n} 3^i \left(\frac{n}{2}\right)^i = \theta(n^{1.585})$$

d) $T(n) = T(n-1) + T\left(\frac{n}{2}\right) + n$

Recurrence	# of nodes	operation	cost
$T(n)$	$1^0 = 1$	1	1
$T\left(\frac{n}{2}\right)$	$1^1 = 1$	1	1
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$T\left(\frac{n}{2^i}\right)$	$1^i = 1$	1	1

$$\sum_{i=0}^{n-1} (n-1) + \sum_{i=0}^{n-1} T\left(\frac{n-i}{2}\right) = \Theta(2^n)$$

e) $T(n) = 2T\left(\frac{n}{4}\right) + \sqrt{n}$

Recurrence	# of nodes	operation	cost
$T(n)$	$2^0 = 1$	$\sqrt{n}$	$\sqrt{n}$
$T\left(\frac{n}{4}\right)$	$2^1 = 2$	$\sqrt{\frac{n}{4}}$	$\sqrt{\frac{n}{2}}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$T\left(\frac{n}{4^i}\right)$	2^i	$\sqrt{\frac{n}{4^i}}$	$\sqrt{\frac{n}{2^i}}$

$$\sum_{i=0}^{\log_4 n} 2^i \sqrt{\frac{n}{4^i}} = \Theta(\sqrt{n} \log n)$$

$$2) a) T(n) = T\left(\frac{n}{2}\right) + \theta(1)$$

$$a = 1, b = 2, f(n) = \theta(1)$$

$$n^{\log_b a} = \log_2 1 = n^0 = 1$$

$$\theta(1) \text{ vs } \theta(n^0) \quad \text{Case 2}$$

$$= \theta(\log n)$$

$$b) T(n) = 7T(n/3) + n^2 - n$$

$$a = 7, b = 3, f(n) = \theta(n^2)$$

$$n^{\log_b a} = \log_3 7 = \frac{\ln 7}{\ln 3} = 1.7712$$

$$n^2 \text{ vs } n^{1.7712} \quad \text{Case 3}$$

$$= \theta(n^2)$$

$$c) T(n) = 4T(n/2) + n^2 \log n$$

$$a = 4, b = 2, f(n) = n^2 \log n$$

$$n^{\log_b a} = n^{\log_2 4} = n^2$$

$$n^2 \log n = \theta(n^{\log_b a} \log^k n) \quad \text{Case 2}$$

$$= \theta(n^2 \log^2 n)$$

$$d) T(n) = 2T(n/4) + \sqrt{n}$$

$$a = 2, b = 4, f(n) = \sqrt{n} = n^{1/2}$$

$$n^{\log_b a} = n^{\log_4 2} = \frac{\ln 2}{\ln 4} = \frac{1}{2}$$

$$\theta(n^{\log_b a})$$

$$= \theta(\sqrt{n} \log n)$$

$$e) T(n) = 7T\left(\frac{n}{8}\right) + \log n$$

$$a = 7, b = 8, f(n) = \log n$$

$$n^{\log_b a} = n^{\log_8 7} = \frac{\ln 7}{\ln 8} = 0.9359$$

$$\log n \text{ vs } n^{0.9359}$$

$$\text{Case 1}$$

$$= \theta(n^{\log_b a})$$